

Vansh Malhotra

AI Engineer | Multi-Agent Systems | LLM Infrastructure | Microservices

✉ vanshmalhotra353@gmail.com 📞 +91 9925351328 📧 vansh-malhotra353 🔄 Vansh3503

SUMMARY

AI Engineer with ~1.5 year of experience building AI-powered systems for QA automation at eInfoChips. Started the AI for QA initiative from scratch and grew it into an 11-member team. Focuses on multi-agent systems, RAG pipelines, and microservice backends. Improved automated test generation accuracy by **60–70%**. Also works on LLM evaluation (DeepEval) and security testing (Garak).

EDUCATION

NEW LJIET (Affiliated with GTU) | GPA: 3.7/4.0
B.E. in Information Technology

Nov 2020 – May 2024
Ahmedabad, India

TECHNICAL SKILLS

Languages	Python, C/C++
AI Frameworks	Agno, CrewAI, LangChain, LangGraph, Langflow, HuggingFace, AutoGen, MCP, A2A
Backend & APIs	FastAPI, SQLAlchemy, Pydantic, REST, WebSockets
Infra & DevOps	Docker, Kubernetes (AKS), Helm, Azure ACR, Azure DevOps, Bitbucket Pipelines, GitHub
Databases	PostgreSQL 16, PgVector, MongoDB, Redis 7, RabbitMQ 3.13
AI/ML Concepts	RAG, Prompt Engineering, Fine-Tuning, Multi-Agent Systems, Multimodal AI
LLM Tooling	LangSmith, DeepEval, Garak, Playwright, Azure OpenAI, Gemini, Ollama
Observability	Kong API Gateway, Structured Logging, Token-Usage Tracking, Tableau

EXPERIENCE

AI Engineer <i>eInfoChips (An Arrow Company)</i>	July 2024 – Present Ahmedabad, India
- Founded the AI Studio initiative at eInfoChips from scratch; it has since grown into an 11-member team building a 11+ microservice platform on Azure AKS with CI/CD via Azure DevOps and Bitbucket Pipelines. - Built a multi-agent QA pipeline using Agno, CrewAI, and LangGraph to auto-generate user stories and test cases from browser recordings and documents — improving test generation accuracy by 60–70%. - Designed an async document processing pipeline: upload → text extraction (Docling/pypdf) → chunking → embedding (Azure OpenAI) → PgVector storage, powering RAG-based Q&A across project knowledge bases. - Built a browser automation microservice using Browser-Use SDK with RabbitMQ task queuing, Redis distributed locking, and live browser streaming via noVNC for AI-driven test execution on AKS. - Developed an internal AI assessment platform for company-wide AI upskilling: FastAPI + React + Redis exam engine with auto-grading and WeasyPrint PDF certificate generation. - Set up LLM evaluation with DeepEval (precision, faithfulness, relevancy) and ran adversarial security probes with Garak (prompt injection, jailbreaks, OWASP LLM Top-10) on Azure OpenAI models. - Built a multimodal RAG system using LLaVA and CLIP for image/video understanding, enabling document Q&A over mixed-media content. - Implemented platform security: Fernet-encrypted API keys, JWT blacklisting in Redis, bcrypt password hashing, Kong rate limiting, and PostgreSQL advisory locks.	
AI Engineer Intern <i>Seaflex Technologies</i>	Jan 2024 – Jul 2024 Ahmedabad, India
Built RESTful backend services for an e-commerce platform using FastAPI — covering product catalogue, cart management, and async order processing.	

PROJECTS

AI Studio — Enterprise AI & QA Platform (<i>eInfoChips</i>)	2024 – Present
11+ microservice platform (Auth, Document, Chat, Workflow, Web-Stories, Web-Testcase, Test-Management, Web-Crawler, and more) actively growing with new capabilities each sprint. Stack: FastAPI · React 18 · PostgreSQL · PgVector · Redis · RabbitMQ · Kong · Docker · AKS · Helm · Agno · CrewAI · LangGraph · Azure OpenAI 60–70% uplift in test generation accuracy and ~50% reduction in QA cycle time.	
LLM Reliability & Security Testing Framework (<i>eInfoChips</i>)	2024 – Present
RAG evaluation with DeepEval; adversarial probing with Garak covering OWASP LLM Top-10 attack categories on Azure OpenAI-hosted models.	

Multimodal RAG — Unified Q&A Platform (*eInfoChips*)

2024

- RAG pipeline over PDF, CSV, DOCX, images, and video using LLaVA (Ollama) + LangChain + FAISS; sub-2s P95 latency for document-grounded answers.

DocuQuery: Intelligent Document Q&A (*Personal*)

2023

- RAG app for Q&A over PDFs, TXT, and URLs using OpenAI, LLaMA 3, Groq, FAISS, and Streamlit.

CERTIFICATIONS

- Oracle Cloud Infrastructure 2024 Generative AI Certified Professional
- IBM — Introduction to Data Analytics
- IBM — Python for Data Science, AI & Development
- IBM — Data Visualization and Dashboards with Excel and Cognos